

On why India could not test a thorium-uranium fuel in existing reactors, as done by CCTE, Kakodkar says there were some experiments, but they lacked the Advanced Heavy Water Reactors to increase the fuel burn-up.

“These are technology developments, and we should be strong in technology development for self-reliance. But if we cannot do that within the time frame, the best way is to collaborate. For Kudankulam, we are importing enriched uranium because we cannot make it today; maybe tomorrow we will be able to make it. The effort will be to continuously improve value addition in the country,” he says.

India is heavily dependent on imports for uranium, and the demand is expected to increase significantly, in line with nuclear capacity expansion, from 1,800–2,000 tonnes/year currently to over 5,000 tonnes/year by 2047.

“Growth will be driven by the target of 100 GW nuclear capacity, implying multi-fold increase in fuel requirements. Given the limited domestic production, incremental demand will be largely met through imports, increasing reliance on long-term international supply agreements,” says Vishal Periwal, Equity Analyst, Infrastructure and Utilities, Prabhudas Lildhar.

Innovations in the nuclear sector have caught the attention of private players. The introduction of the SHANTI Act has opened the doors of the sector to private players.

India’s nuclear programme+ has historically been entirely government funded. Marking a structural shift in India’s nuclear policy framework, the SHANTI Act, 2025 (Sustainable Harness-

ing and Advancement of Nuclear Energy for Transforming India, 2025) has opened a sector for regulated private sector participation in nuclear and fusion energy under Atomic Energy Regulatory Board (AERB) oversight. Big names such as Reliance and Adani have held discussions on investments in future technologies that require neither uranium nor plutonium as fuel nor create long-lived radioactive waste. The aim is to build a cleaner and safer

FUSION PUSH

Another solution is fusion-based reactor technology. It is gaining traction with a few Indian start-ups looking at power generation by 2035. Pranos, Anubal Fusion, Hylene Technologies, and ASPL Fusion are among the key players looking to make India energy independent using this yet-to-be-proven commercial energy source. The good part, all of them are working on different technologies to make a breakthrough



“India must look for different nuclear technologies [other than the three-stage programme] to achieve the target of 100GW of nuclear capacity by 2047”

ANIL KAKODKAR

FORMER CHAIRMAN, ATOMIC ENERGY COMMISSION



alternative to nuclear fission reactors in India. Cumulative venture capital in fusion energy companies has reached about \$11 billion, according to the International Energy Agency’s 2026 report.

Additionally, private players are already in talks with the Nuclear Power Corporation of India Ltd (NPPCIL) for investment in Bharat Modular Reactors and Bharat Small Modular Reactors, providing captive power to industries. Larson & Toubro has already tied up with the US-based Holtec International for deployment of its Small Modular Reactors (SMRs).

in the commercial deployment of fusion-based reactors. Pranos Fusion raised ₹63 crore in March this year from VC funds, and is in talks for partnerships with private players.

India has fission-based reactors, which split a larger atom (Uranium) into several smaller atoms to release energy. Fusion is the opposite of fission; you combine two smaller atoms to form a larger atom, and in the process, release energy that is exponentially higher than what is given out in a fission reaction. The most common fusion reactions